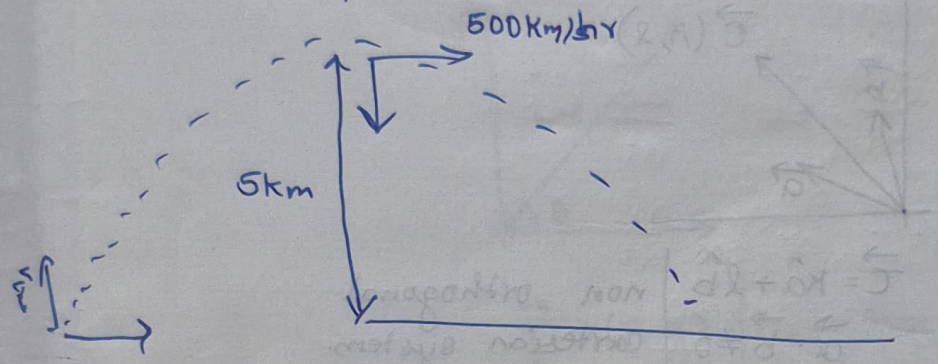
Intermediate Mechanics I

A. An aircraft flies at a speed of 500 Km/hr. It drops a bomb at a given point in time. How far will the bomb fall from the point. If the height of air craft is 5 Km vertically.

$h = \cancel{2Km}\ 5Km$

$s = 500Km/h$



$t =$

$s = ut + \frac{1}{2}at^2$

$5Km = -\frac{1}{2} \times 9.8 \times t^2$

$\sqrt{\frac{5000 \times 2}{9.8}} = t$

$31.94 sec = t$

$s = \frac{d}{t}$

$d = st$

$d = 138.8 \times 31.9$

$d = 4430m$

$= 4.43Km$

B. A wheel rotating at 25 rev/sec undergoes a deacc. of 5 rad/sec^2. How many rev will it make before cmg to stop?

$25 rev/sec = 50\pi rad/sec$

$\cancel{25} - 50\pi^2 = -2 \times 5 \times s$

$\frac{50 \times 5 \cancel{0} \pi^2}{\cancel{10}} = s$

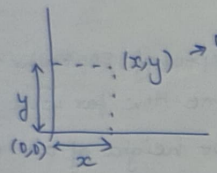
$250\pi^2 = s$

$\frac{\cancel{250}\pi^2}{\cancel{2\pi}} = rev$

$rev = 125\pi$

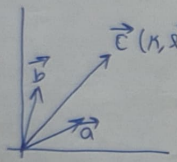
$rev = 392.5$

2-D



→ orthonormal cartesian system

measure distance = Cartesian System



$\vec{c} = \kappa\hat{a} + \ell\hat{b}$ | non orthogonal cartesian system.
$\vec{a}\cdot\vec{b} \neq 0$ |

$\vec{d} = m\hat{a} + n\hat{b}$

$\vec{c}\cdot\vec{d} = \kappa m + \ell n + m\ell\,\hat{a}\cdot\hat{b} + \kappa n\,\hat{a}\cdot\hat{b}$

$\Rightarrow \vec{c}\cdot\vec{d} = \kappa m + \kappa n\hat{a}\cdot\hat{b} + \ell n + \ell m\,\hat{b}\cdot\hat{a}$

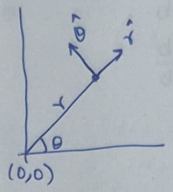
3-D

$\hat{i}\cdot\hat{i} = 1 \quad \hat{i}\times\hat{i} = 0$

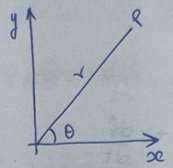
$\hat{i}\times\hat{j} = \hat{k}$ right handed orthonormal cartesian system.

$\hat{i}\times\hat{j} = -\hat{k}$ left handed orthonormal cartesian system.

Plane Polar Coordinates : orthonormal system.

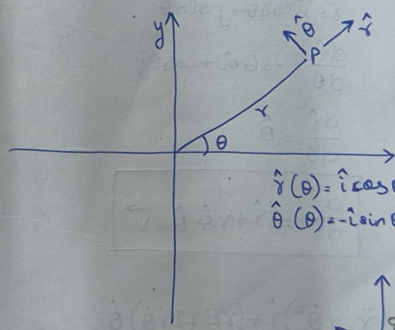


$\hat{r}\cdot\hat{r} = 1$
$\hat{\theta}\cdot\hat{\theta} = 1$



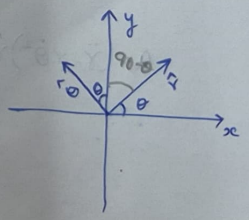
$\vec{r} = r(x,y)$	inverse
$\vec{r} = r(r,\theta)$	$r = \sqrt{x^2+y^2}$
$x = r\cos\theta$	$\theta = \tan^{-1}\left(\frac{y}{x}\right)$
$y = r\sin\theta$	

Unit Vectors in PP Coordinates



$\hat{r}(\theta) = \hat{i}\cos\theta + \hat{j}\sin\theta$

$\hat{\theta}(\theta) = -\hat{i}\sin\theta + \hat{j}\cos\theta$



eg: $r_1(5m, 45°)$

$r_2(5m, 225°)$

$\vec{r} = r\hat{r}$

$r_1 = 5\hat{r}$

$r_2 = 5\hat{r}$

$r_1 + r_2 = 10\hat{r}$

$= 0$

$$\vec{r} = r\hat{r}$$

$$\frac{d\vec{r}}{dt} = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt}$$

$$= \dot{r}\hat{r} + r\frac{d\hat{r}}{dt}$$

$$\frac{d\hat{r}}{dt} = \frac{d\hat{r}}{d\theta}\frac{d\theta}{dt}$$

$$= \frac{d\hat{r}}{d\theta}\dot{\theta}$$

$$\hat{r} = \hat{i}\cos\theta + \hat{j}\sin\theta$$

$$\frac{d\hat{r}}{d\theta} = -\sin\theta\hat{i} + \cos\theta\hat{j}$$

$$\frac{d\hat{r}}{d\theta} = \hat{\theta}$$

$$\boxed{\frac{d\vec{r}}{dt} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} = \vec{v}}$$

$$a = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}$$

Q. Find acceleration

$$\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$$

$$\frac{d\hat{r}}{dt} = \dot{\theta}\hat{\theta}$$

$$\frac{d\hat{\theta}}{dt} = -\dot{\theta}\hat{r}$$

$$\vec{v} = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt}$$

$$\frac{d\vec{v}}{dt} = \hat{r}\frac{d\dot{r}}{dt} + \frac{dr}{dt}\frac{d\hat{r}}{dt} + \frac{dr}{dt}\frac{d\hat{\theta}}{dt} + \frac{d^2\hat{r}}{dt^2}\frac{dr}{dt}$$

$$\ddot{r}\hat{r} + \dot{r}\dot{\theta}\hat{\theta} + \dot{r}\dot{\theta}\hat{\theta} + \frac{d^2\hat{r}}{dt^2}\dot{r}$$

$$\ddot{r}\hat{r} + 2\dot{r}\dot{\theta}\hat{\theta} + \frac{d^2\hat{r}}{dt^2}\dot{r}$$

$$\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$$

$$\frac{dr\dot{\theta}\hat{\theta}}{dt} = r\dot{\theta}\frac{d\hat{\theta}}{dt} + \frac{dr\dot{\theta}}{dt}\hat{\theta}$$

$$\frac{d\vec{r}\dot{r}\hat{r}}{dt} = \ddot{r}\hat{r} + \dot{r}\frac{d\hat{r}}{dt} = \ddot{r}\hat{r} + \dot{r}\dot{\theta}\hat{\theta}$$

$\frac{dr\dot{\theta}\hat{\theta}}{dt} \Rightarrow r, \dot{\theta}, \hat{\theta}$ depends on time ; So diff each wrt time:

$$= \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} + r\dot{\theta}\frac{d\hat{\theta}}{dt}$$

$$= r\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} - r\dot{\theta}^2\hat{r}$$

$$\frac{dv}{dt} = \ddot{r}\hat{r} + \dot{r}\dot{\theta}\hat{\theta} + \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} - r\dot{\theta}^2\hat{r}$$

$$a = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}$$

Find acceleration

$$\vec{V} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$$

$$\frac{d\vec{V}}{dt} = \frac{d}{dt}\left(\dot{r}\hat{r} + r\dot{\theta}\hat{\theta}\right)$$

$$= \frac{d\hat{r}}{dt}\dot{r} + \dot{r}\frac{d\hat{r}}{dt} + \frac{d(r\dot{\theta})}{dt}\hat{\theta} + (r\dot{\theta})\frac{d\hat{\theta}}{dt}$$

$$= \ddot{r}\hat{r} + \dot{r}\frac{d\hat{r}}{dt} + (\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta} + r\dot{\theta}\frac{d\hat{\theta}}{dt}$$

$$\frac{d\vec{r}}{dt} = \dot{\theta}\hat{\theta} \qquad \frac{d\hat{\theta}}{dt} = -\dot{\theta}\hat{r}$$

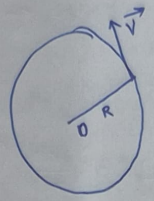
$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}$$

$$\vec{V} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$$

$$\hat{r} = i\cos\theta + j\sin\theta$$

$$\hat{\theta} = -i\sin\theta + j\cos\theta$$

Uniform Circular Motion



$$\vec{V} = 0 + r\omega\hat{\theta}$$

$$\vec{a} = (0 - r\omega^2)\hat{r} + (0+0)\hat{\theta}$$

$$\boxed{a = -\omega^2 r\hat{r}}$$

Example: r = constant

$$\dot{\theta} = ct$$

$$\ddot{\theta} = c$$

$$\vec{V} = 0 + rct\hat{\theta}$$

$$\vec{a} = (-rc^2t^2)\hat{r} + (rc)\hat{\theta}$$

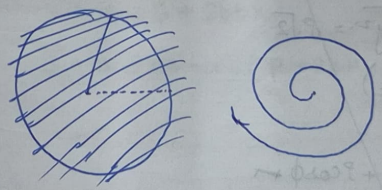
$$\vec{a} = -r(ct)^2\hat{r} + cr\hat{\theta}$$

Q. A disc rotates at constant angular speed ω. An ant ~~is~~ crawls directly towards the rim at a constant speed 'b'.

a) Plot the motion of ant?

b) Find $\vec{V}$ and $\vec{a}$?

a.



b.

$$r = bt + r_0 \qquad \theta = \omega t + \omega_0$$

$$\dot{r} = \frac{dr}{dt} = b \qquad \dot{\theta} = \frac{d\theta}{dt} = \omega$$

$$\ddot{r} = 0 \qquad \ddot{\theta} = 0$$

$$V = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$$

$$= b\hat{r} + (bt + r_0)\omega\hat{\theta}$$

$$a = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}$$

$$(0 - (bt + r_0)\omega^2)\hat{r} + (2b\omega)\hat{\theta}$$

$$a = (-(bt\omega^2 + r_0\omega^2))\hat{r} + 2b\omega\hat{\theta}.$$

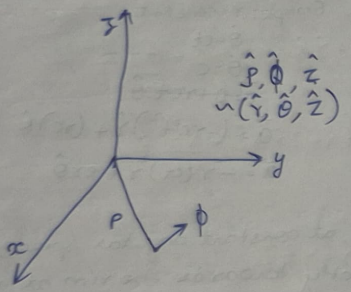
Kinematics in 3D

$r = x\hat{i} + y\hat{j} + z\hat{k}$

$v = \dot{x}\ \ \dot{y}\ \ \dot{z}$

$a = \ddot{x}\ \ \ddot{y}\ \ \ddot{z}$

Cylindrical Coordinates



eg: $x = \rho\cos\phi$

$y = \rho\sin\phi$

$z = z$

$\rho = \sqrt{x^2+y^2} = \rho\sqrt{2}$

$\phi = \tan^{-1}\left(\frac{y}{x}\right)$

$z = z$

$\vec{v} = -\rho\sin\phi + \rho\cos\phi$

$\vec{a} = -\rho\cos\phi - \rho\sin\phi$

eg: $\rho\cos\phi = x$

$\rho\sin\phi = y$

$z = z$

$\rho = \sqrt{x^2+y^2}$, $\phi = \tan^{-1}\left(\frac{y}{x}\right)$, $z = z$

$\vec{v} = \dot{\rho}\hat{\rho} + \rho\dot{\phi}\hat{\phi} + \dot{z}\hat{k}$

$\vec{a} = (\ddot{\rho} - \rho\dot{\phi}^2)\hat{\rho} + (\rho\ddot{\phi} + 2\dot{\rho}\dot{\phi})\hat{\phi} + \ddot{z}\hat{k}$

8. A bead slides on a wire bent into a helix. The position is given as

$\rho = a$

$\phi = \omega t$

$z = bt^2$

find $\vec{v}t, \vec{a}t$

~~$\vec{r} = a\hat{\theta}$~~

$\dot{\rho} = 0 \quad \dot{\phi} = \omega \quad \dot{z} = 2bt$

$\ddot{\rho} = 0 \quad \ddot{\phi} = 0 \quad \ddot{z} = 2b$

$\vec{v} = a\omega\hat{\phi} + 2bt\hat{k}$

$\vec{a} = ~~2b\hat{k}~~ - a\omega^2\hat{\rho} + 2b\hat{k}$

Spherical Polar Coordinates

r, θ, ϕ

$x = r\sin\theta\cos\phi$

$y = r\sin\theta\sin\phi$

$z = r\cos\theta$

$r = (x^2+y^2+z^2)^{1/2}$

* $\frac{y}{x} = \frac{\sin\phi}{\cos\phi}$

$\frac{y}{x} = \tan\phi$

$\phi = \tan^{-1}\left(\frac{y}{x}\right)$

* $x^2+y^2 = r^2\sin^2\theta\cos^2\phi + r^2\sin^2\theta\sin^2\phi$

$x^2+y^2 = r^2\sin^2\theta(\cos^2\phi+\sin^2\phi)$

$\sqrt{x^2+y^2} = r\sin\theta$

divide by $r\cos\theta$ ie z

$\frac{\sqrt{x^2+y^2}}{z} = \tan\theta$

$$\boxed{\theta = \tan^{-1}\left(\frac{\sqrt{x^2+y^2}}{z}\right)}$$

Now the other way

$\theta = \tan^{-1}\left(\frac{\sqrt{x^2+y^2}}{z}\right)$

$\phi = \tan^{-1}\left(\frac{y}{x}\right)$

$\tan\phi = \frac{y}{x}$

$\frac{\sin\phi}{\cos\phi} = \frac{y}{x}$

~~$x\sin\phi = y\cos\phi$~~

~~multiply $\cos\phi$ & x on numerator & denominator~~

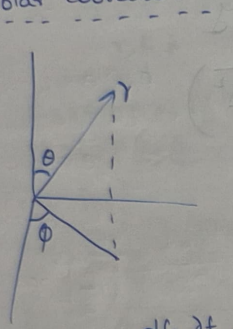
~~$x\sin\phi = y\cos\phi$~~

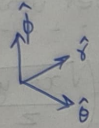
$\tan\theta = \frac{\sqrt{x^2+y^2}}{z}$

$\tan^2\theta = \frac{x^2+y^2}{z^2}$

$\frac{\sin^2\theta}{\cos^2\theta} = \frac{x^2+y^2}{z^2}$

Spherical Polar Coordinates





$\hat{r}(\theta,\phi)$

$\hat{\theta}(\theta,\phi)$

$\hat{\phi}(\phi)$

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz$$

Assignment 1: derivation of velocity in plane polar coordinates.

$$\frac{d\hat{r}}{dt} = \vec{V} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\dot{\phi}\hat{\phi}$$

$\hat{r} = \sin\theta\cos\phi\hat{i} + \sin\theta\sin\phi\hat{j} + \cos\theta\hat{k}$

$\hat{\theta} = \cos\theta\cos\phi\hat{i} + \cos\theta\sin\phi\hat{j} - \sin\theta\hat{k}$

$\hat{\phi} = -\sin\theta\hat{i} + \cos\phi\hat{j}$

a) ST they are mutually orthogonal

b) $\hat{r}\times\hat{\theta}$

i	j	k
$\sin\theta\cos\phi$	$\sin\theta\sin\phi$	$\cos\theta$
$\cos\theta\cos\phi$	$\cos\theta\sin\phi$	$-\sin\theta$

$$\hat{i}\left(\sin^2\theta\sin\phi - \cos^2\theta\sin\phi\right) - \hat{j}\left(-\sin^2\theta\cos\phi - \cos^2\theta\cos\phi\right)$$

$$+\hat{k}\left(\sin\theta\sin\phi\cos\theta\cos\phi - \sin\theta\sin\phi\cos\theta\cos\phi\right)$$

$$\hat{i}\left(\sin\phi\left(\sin^2\theta - \cos^2\theta\right)\right) + \hat{j}\left(\cos\phi\left(\sin^2\theta + \cos^2\theta\right)\right)$$

a) $\hat{r}\cdot\hat{\theta} = \sin\theta\cos^2\phi\cos\theta + \sin\theta\cos\theta\sin^2\theta - \sin\theta\cos\theta$

$\sin\theta\cos\theta(1-1) = 0$

$\hat{\theta}\cdot\hat{\phi} = -\sin\theta\cos\theta\cos\phi + \cos\theta\cos\phi\sin\phi$

$\hat{r}\cdot\hat{\phi} = -\sin^2\theta\cos\phi + \cos\phi\sin\theta\sin\phi$

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2 - r\dot{\phi}^2\sin^2\theta)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\dot{\phi}^2\sin\theta\cos\theta)\hat{\theta}$$

$$+(r\ddot{\phi}\sin\theta + 2\dot{r}\dot{\phi}\sin\theta + 2r\dot{\theta}\dot{\phi}\cos\theta)\hat{\phi}$$

A partical is at (3,4,5) find r, θ, ϕ

$3 = r\sin\theta\cos\phi$

$4 = r\sin\theta\sin\phi$

$5 = r\cos\theta$

$\phi = \tan^{-1}\left(\frac{y}{x}\right)$

$\phi = \tan^{-1}\left(\frac{4}{3}\right)$

$\theta = \tan^{-1}\left(\frac{\sqrt{x^2+y^2}}{z}\right)$

$\theta = \tan^{-1}\left(\frac{\sqrt{9+16}}{5}\right)$

$\theta = \tan^{-1} 1$

$\theta = \frac{\pi}{4}$

$r = \sqrt{3^2+4^2+5^2}$

$r = \sqrt{9+16+25}$

$r = \sqrt{50} = 7.07$

Why the body moves?

Force

$F \rightarrow$ Constant

$F(x) = -kx, \sim \frac{GMm}{r^2}$

$F(t) = \sim e^{at}, bt$

$F(v) = \sim$ drag

$F(t)$

$a(t) = \frac{F}{m}$

$\frac{dv}{dt} = \frac{F}{m}$

$\int dv = \int \frac{F}{m}\,dt$

$v(t) =$

$\frac{dx}{dt} = v(t)$

① A force $F = 3t^2$ N acts on a body for 3 sec after which the partical moves freely. The partical is initially at rest at the origin. Find its position at $t = 5$ sec. (mass = 0.1 kg)

$\int dv = \int_0^3 \frac{3t^2\,dt}{0.1}$

$v(t) = \left[t^3\right]_0^3 \times 10$

$v(t) = 270$ m/s

$\frac{dx}{dt} = t^3$

$x = \frac{10t^4}{4}$

$= \frac{810}{4}$ m $= 202.5$ m

V will be cst after $t = 3$

$270 \times 2 = 540$ m

total $= 540 + 202.5 =$ 742.5 m

$s = ut + \frac{1}{2}at^2$

$s = 270 \times 2 = 540 + \frac{81}{4}$

$\frac{2160+81}{4} = \frac{2241}{4}$ m $= 560.25$ m

560.25

540 × 4 = 2160, 2160 + 81 = 2241

particle of mass m is acted upon by force $F(t) = Ae$ initially at rest at $x = 0$ find $v(t)$ & $x(t)$ at $t = \infty$

$$F(t) = Ae^{-bt}$$

$$a = \frac{F}{m}$$

$$\frac{dv}{dt} = \frac{Ae^{-bt}}{m}$$

$$dv = \frac{Ae^{-bt}}{m}\,dt$$

$$v(t) = -\frac{Ae^{-bt}}{mb}$$

$$\frac{dx}{dt} = \frac{-Ae^{-bt}}{mb} = 0$$

$$dx = \frac{-Ae^{-bt}}{mb}\,dt$$

$$x(t) = \frac{Ae^{-bt}}{mb^2} = 0$$

$$[v(t)]_0^\infty = \frac{A}{mb}$$

~~$[x(t)]_0^\infty = -\frac{A}{mb^2}$~~

Force as a f^n of Velocity

$$F = F(v)$$

$$\frac{dv}{dt} = \frac{F(v)}{m}$$

$$\frac{dv}{F(v)} = dt$$

$$\frac{m\,dv}{F(v)} = dt$$

$$\int \frac{dv}{F(v)} = \int \frac{dt}{m}$$

$$\frac{t}{m} = \int \frac{dv}{F(v)}$$

↳ t in terms of $t(v)$
invert to get $v(t)$
then find
$x = \int v\,dt$

Q.) A boat is being rowed towards shore. Just before reaching the shore, rowing stops & resistance of the water brings it to stop. The resistance is modelled as $F = -bv$. Find $x(t)$?

$$\frac{dv}{dt} = -\frac{bv}{m}$$

$$\int \frac{dv}{bv} = -\frac{t}{m}$$

$$\frac{1}{b}\ln v = -\frac{t}{m}$$

$$t = -\frac{m}{b}\ln v$$

$$-\frac{bt}{m} = \ln v$$

$$e^{-\frac{bt}{m}} = v$$

$$x = \int e^{-bt/m}$$

$$-\frac{b}{m}e^{-bt/m}\left[-\frac{m}{b}e^{-bt/m}\right]$$

Cont.

$$v(t) = v_0 e^{-bt/m}$$

Integrating

$$\int_{x_0}^{x_t} dx = \int_0^t v_0 e^{-bt/m}\, dt$$

$$x(t) - x(0) = \left[\frac{-v_0 m}{b} e^{-bt/m}\right]_0^t$$

$$x(t) = x(0)\left[\frac{-v_0 m}{b} e^{-bt/m} + \frac{v_0 m}{b} e^{0}\right]$$

$$x(t) = \boxed{x(0) + \frac{v_0 m}{b}\left[1 - e^{-bt/m}\right]}$$

What is x when $t \to \infty$

$x = x(\infty) = x_0 + \frac{v_0 m}{b}$

as $e^{-\infty}$ becomes very close to 0

What is v when $t \to \infty$

↳ becomes very small even at $t \to \infty$.

Various Drag Forces

* If velocity ~~<20m/s~~ >20 m/s

$F = -Dv^2$ $\quad F \propto v^2$

$D = \frac{1}{2} C_D A \rho$

C_D: drag coeff (≈ 0.2)

A = Cross section area

ρ = Air ~~resis~~ density.

* When velocity < 20 m/s

$F \propto v$

* When velocity ≈ speed of sound.

different model.

Force as a Function of position

$F = F(x)$

$$m\frac{dv}{dt} = F(x)$$

$$m\frac{dv}{dx}\frac{dx}{dt} = F(x)$$

$$m\frac{dv}{dx} v = F$$

$$mv\,dv = F\,dx$$

integrate

$$m\int_{v_0}^{v} v\,dv = \int F\,dx$$

$$\boxed{\frac{m}{2}\left[v^2 - v_0^2\right] = \int F \cdot dx}$$

$F = -kx$

$x = A\sin(\omega t + \phi)$

Example 1: A particle of mass 'm' has a velocity $v = A\cos\alpha x$ A & α are constant.

Find $F(x)$

$$m\frac{dv}{dx}\frac{dx}{dt} = Fx$$

$$\frac{dv}{dx} = -A\alpha\sin\alpha x$$

$$m(-A\alpha\sin\alpha x)(A\cos\alpha x) = Fx$$

$$\boxed{F(x) = -A^2 m\alpha \sin\alpha x \cos\alpha x}$$

$$\boxed{F(x) = \frac{-A^2 m\alpha}{2}\sin 2\alpha x}$$

WORK-ENERGY Ch-5

Work done on a particle by a force.

$$W = \int_{r_1}^{r_2} F \cdot ds$$

ds = displacement

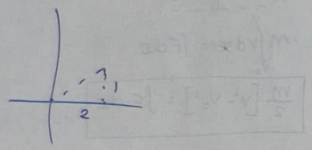
$$W = \int_{r_1}^{r_2} F \cdot ds = \Delta T$$

W = Change in Kinetic energy.

Example: A particle is acted upon by a force

$$F = (ax^3 + bxy^2)\hat{i} + (ay^3 + bx^2y)\hat{j}$$

Calculate the work done by this force from (0,0) to (2,1) along
a) a straight line



$2t \le t \le 1$

$x = 2t \quad \frac{dx}{dt} = 2$

$y = t$

~~F=~~

~~F ds=~~

$dx = 2dt$

$\frac{dy}{dt} = 1$

$dy = dt$

$$F \cdot ds = (ax^3 + bxy^2)\,dx + (ay^3 + bx^2y)\,dy$$

$$F \cdot ds = 2(8at^3 + 2bt^3)dt + (at^3 + 4bt^3)dt$$

$$\int_0^1 F \cdot ds = \int_0^1 16at^3 + 4bt^3 + at^3 + 4bt^3\,dt$$

$$\int_0^1 17at^3 + 8bt^3\,dt$$

$$\left[\frac{17at^4}{4} + \frac{8bt^4}{4}\right]_0^1 = \frac{17a}{4} + 2b$$

b) along first x & y.

A force is expressed as $F = xy\hat{i} + x^2\hat{j}$. Find the work done by the force from (0,0) to (2,4) along path a) a straight line
b) a parabola.

a) $x = 2t$, $y = 4t$, $0 \le t \le 1$

$\frac{dx}{dt} = 2$, $\frac{dy}{dt} = 4$

$8t^2 t + 4t^2 j$

$\int_0^1 16t^2 + 16t^2$

$\left[\frac{16t^3}{3} + \frac{16t^3}{3}\right]_0^1$

$\frac{16}{3} + \frac{16}{3} = \frac{32}{3} J$

b) $x = t$, $y = t^2$, $0 \le t \le 2$

$\frac{dx}{dt} = 1$, $\frac{dy}{dt} = 2t$

$F = t^3 \frac{dx}{dt} + t^2 \frac{dy}{dt}$

$\int_0^2 t^3 + 2t^3$

$\left[\frac{t^4}{4} + \frac{2t^4}{2}\right]_0^2$

$4 + 8 = 12 J$

Scalar, Vector Fields

$x, y, z \rightarrow f \rightarrow$ scalar field

$x, y, z \rightarrow \vec{A} \rightarrow$ vector field

$\vec{\nabla} \rightarrow i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}$

∇f where f is a scalar field

$\rightarrow \hat{i}\frac{\partial f}{\partial x} + \hat{j}\frac{\partial f}{\partial y} + \hat{z}\frac{\partial f}{\partial z}$

this converts a scalar to a vector field.

example: Evaluate the gradient of $f = 5x^3 + 6y^2 + 2z$ at 1, 2, 3

$\nabla f = 15x^2\hat{i} + 12y\hat{j} + 2\hat{k}$

$= 15\hat{i} + 24\hat{j} + 2\hat{k}$

Divergence $= \vec{\nabla} \cdot \vec{A} \rightarrow$ scalar field

Curl $= \vec{\nabla} \times \vec{A} \rightarrow$ vector field

example: find the curl and divergence of the f^n $\vec{F} = xy\hat{i} + x^2\hat{j} + 0\hat{k}$

~~$\vec{\nabla} = y\hat{i}$~~

a) $\vec{\nabla} \times \vec{A}$

$$\begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & x^2 & 0 \end{vmatrix}$$

$2x - x = x\hat{k}$

b) $\frac{\partial}{\partial x} xy\hat{i} + \frac{\partial}{\partial y} x^2 + \frac{\partial}{\partial z} 0$

$= y$

Property of Gradient

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy + \frac{\partial u}{\partial z}dz$$

$$ds = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$$

$$\nabla u = i\frac{\partial u}{\partial x} + j\frac{\partial u}{\partial y} + k\frac{\partial u}{\partial z}$$

$$\boxed{du = \nabla u \cdot ds}$$

If you integrate

$$\int du = \int \vec{\nabla u}\cdot ds$$

→ this is path independent!

$$= u_2 - u_1$$

Del in other Representation

→ Coordinate transformations

$$x = x(q_1, q_2, q_3)$$
$$y = y(q_1, q_2, q_3)$$
$$z = z(q_1, q_2, q_3)$$

→ Inverse

$$q_1 = q_1(x, y, z)$$
$$q_2 = q_2(x, y, z)$$
$$q_3 = q_3(x, y, z)$$

$$\vec{\nabla} = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$$

$\vec{\nabla} f = \vec{A}$

$f\vec{\nabla} = ?$

will give out another coordinate.

$$\vec{ds} = i\,dx + j\,dy + k\,dz$$

$ds^2 = dx^2 + dy^2 + dz^2$ is an invariant

maxima

$$x = x(q_1, q_2, q_3)$$

$$(dx)^2 = \left(\frac{\partial x}{\partial q_1}dq_1 + \frac{\partial x}{\partial q_2}dq_2 + \frac{\partial x}{\partial q_3}dq_3\right)^2$$

$$(dy)^2 = \left(\quad\right)^2$$

$$(dz)^2 = \left(\quad\right)^2$$

$$ds^2 = \sum_{ij} h_{ij}^2\, dq_i\, dq_j \;;\; i,j = 1,2,3$$

h_{ij}^2 = scale factor

9 terms in this case. $\Rightarrow h_{ij}$: metric tensor or metric.

→ If the coordinate system is orthogonal

$$h_{ij} = 0 \text{ if } i \neq j$$

$$ds^2 = h_{11}dq_1^2 + h_{22}dq_2^2 + h_{33}dq_3^2$$

$h_{11} \to h_1$, $h_{22} \to h_2$, $h_{33} \to h_3$

$$ds = h_1 dq_1$$

$$\nabla = i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}$$

$$= \hat{r}\frac{\partial}{h_1 dr} + \hat{\theta}\frac{\partial}{h_2 d\theta} + \hat{\phi}\frac{\partial}{h_3 d\phi}$$

Find $\vec{\nabla}$ in cylindrical coordinates.

$$x = \rho\cos\phi$$
$$y = \rho\sin\phi$$
$$z = z$$

$$x^2 + y^2 = \rho^2$$
$$\rho = \sqrt{x^2 + y^2}$$

$$(dx)^2 = (\rho\sin\phi\, d\phi + d\rho\cos\phi)^2$$
$$(dy)^2 = (\rho\cos\phi\, d\phi + d\rho\sin\phi)^2$$
$$(dz)^2 = (dz)^2$$

$$dx^2 + dy^2 + dz^2 = d^2\rho\cos^2\phi + \rho^2\sin^2\phi\, d^2\phi - 2\sin\phi\,\rho\cos\phi\, d\rho\, d\phi$$
$$+ \rho^2\cos^2\phi\, d^2\phi + d^2\rho\sin^2\phi + 2\rho\cos\phi\, d\rho\sin\phi\, d\phi + d^2z$$
$$d^2\rho(1) + \rho^2 d^2\phi(1) + d^2z$$

$$1\, d^2\rho + \rho^2 d^2\phi + d^2z$$
$$h_1^2 \quad h_2^2 \quad h_3^2$$

$$\vec{\nabla} = \frac{1}{h_1}\hat{\rho}\frac{\partial}{\partial\rho} + \frac{1}{h_2}\hat{\phi}\frac{\partial}{\partial\phi} + \frac{1}{h_3}\hat{z}\frac{\partial}{\partial z}$$

$$\vec{\nabla} = \hat{\rho}\frac{\partial}{\partial\rho} + \frac{1}{\rho}\hat{\phi}\frac{\partial}{\partial\phi} + \frac{1}{1}\hat{z}\frac{\partial}{\partial z}$$

Find $\vec{\nabla}$ in spherical coordinates.

$$x = r\sin\theta\cos\phi$$
$$y = r\sin\theta\sin\phi$$
$$z = r\cos\theta$$

~~$dx = dr\cos\theta$~~

$$dx = dr\sin\theta\cos\phi + r\cos\theta\cos\phi\, d\theta - r\sin\theta\sin\phi\, d\phi$$
$$dy = dr\sin\theta\sin\phi + r\cos\theta\sin\phi\, d\theta + r\sin\theta\cos\phi\, d\phi$$
$$dz = dr\cos\theta - r\sin\theta\, d\theta$$

~~$d^2x =$~~

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$$
$$d^2x = d^2r^2\sin^2\theta\cos^2\phi + r^2\cos^2\theta\cos^2\phi\, d\theta^2 + r^2\sin^2\theta\sin^2\phi\, d\phi^2$$
$$d^2y = dr^2\sin^2\theta\sin^2\phi + r^2\cos^2\theta\sin^2\phi\, d\theta^2 + r^2\sin^2\theta\cos^2\phi\, d\phi^2$$
$$d^2z = dr^2\cos^2\theta + r^2\sin^2\theta\, d^2\theta - 2r\sin\theta\cos\theta\, dr\, d\theta$$

$$d^2x + d^2y + d^2z = dr^2\left(\sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi + \cos^2\theta\right)$$

$$W = \int_{Y_1, C}^{Y_2} F \cdot ds = \Delta T$$

For Conservative forces, when $\nabla \times F = 0$

we can write:

$$F = \pm \nabla V$$

Since $\nabla \times \nabla V = 0$

we choose

$$F = -\nabla V$$

$$W_g = -\nabla V$$

$$\Delta T = -\nabla V$$

$$T_2 - T_1 = -(V_2 - V_1)$$

hence, $V_1 + T_1 = V_2 + T_2$

Q. A particle has a PE given by $V = V(x) = 3x^2 e^x$ what is the force on x-particle a) when $x = 0m$?

b) when $x = 2m$?

What is the work done by the force from $x = 0$ to $x = 2m$.

Is the force conservative? ⇒ Yes.

$$\nabla V = 6xe^x + 3x^2 e^x \hat{i}$$

$$F = -\nabla V = (-6xe^x - 3x^2 e^x)\hat{i}$$

$$F(0) = 0$$

$$F(2) = -e^2 (12 + 12)$$

$$= 24x - e^2$$

$$= -177.3N$$

$$W = -\int_0^2 (6xe^x + 3x^2 e^x)\, dx$$

$$-\left[\int_0^2 6xe^x\, dx + \int_0^2 3x^2 e^x\, dx\right]$$

$$W = -(V(2) - V(0))$$

$$= -(+88.66)$$

$$= -88.66J$$

Energy diagrams



(the peak it gets is an unstable oscillation. b/w the wells it will keep an oscillating.

$$\epsilon = T + V$$

$$T = \epsilon - V$$

$$\frac{1}{2} m v^2 = \epsilon - V$$

$$v^2 = \frac{2}{m}(\epsilon - V)$$

$$v = \sqrt{\frac{2}{m}(\epsilon - V)}$$

$$\frac{dx}{dt} = \sqrt{\frac{2}{m}(\epsilon - V)}$$

$$\int \frac{dx}{\sqrt{\frac{2}{m}(\epsilon - V)}} = \int dt$$

P) A particle of mass $m = 0.1kg$ moves in one dimensional potential given by $V = -2x + 3x^2$ find $x(t)$ if $\epsilon = 1J$

$$\int \frac{dx}{\sqrt{20(2x - 3x^2)}} = \int dt$$

$$\int \frac{dx}{\sqrt{40x - 60x^2}} = \int dt$$

$$\int (40x - 60x^2)^{-1/2}\, dx = \int dt$$

$$\frac{40x - 60x^2}{\frac{1}{2}(40x - 120)}^{1/2} = t$$

$$\int \frac{dx}{\sqrt{a + 2bx - cx^2}} = \frac{1}{\sqrt{c}} \sin^{-1} \frac{cx - b}{\sqrt{b^2 + ac}}$$

$$\int \frac{dx}{\sqrt{20(1 + 2x - 3x^2)}}$$

$$\int \frac{dx}{\sqrt{20 + 40x - 60x^2}} =$$

$$\frac{1}{\sqrt{60}} \sin^{-1} \frac{60x - 20}{\sqrt{20^2 + 1200}}$$

$$\frac{1}{\sqrt{60}} \sin^{-1} \frac{60x - 20}{40} = \frac{1}{\sqrt{60}} \sin^{-1} \frac{(60x - 20)}{\sqrt{1600}} = \frac{1}{\sqrt{60}} \sin^{-1} \frac{60x - 20}{40}$$

$$t = \frac{1}{\sqrt{60}} \sin^{-1}\left(\frac{3x-1}{2}\right) + C$$

$$\sqrt{60}(t+C) = \sin^{-1}\left(\frac{3x-1}{2}\right)$$

$$\sin\left(\sqrt{60}(t+C)\right) = \frac{3x-1}{2}$$

$$\frac{2\sin\left(\sqrt{60}t + \sqrt{60}C\right)}{3} = 3x-1$$

~~3x+1~~

$$\frac{2\sin\left(\sqrt{60}(t+C)\right)+1}{3} = x$$

$$\boxed{\frac{2}{3}\sin\left(\sqrt{60}(t+C)\right) + \frac{1}{3} = x(t)}$$

Pr.2) Find the COM of system of particle, mass 1,2,3 located at (0,1), (0), (1,1)

$$\frac{\hat{j} + 2\hat{i} + 3\hat{i} + 3\hat{j}}{6}$$

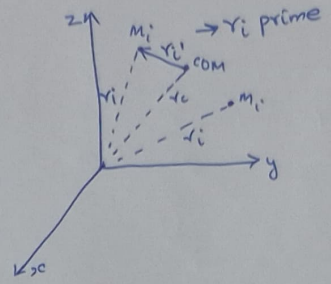
$$= \frac{5\hat{i} + 4\hat{j}}{6}$$

$$\frac{5}{6}\hat{i}, \frac{4}{6}\hat{j}$$

Energy Of a Systems Of particle

COM =

$$r_c = \frac{\sum m_i r_i}{\sum_{i=1}^{N} m_i = M}$$



$$M r_c = \sum m_i r_i$$

$$M \dot{r}_c = \sum m_i \dot{r}_i$$

$$\dot{r}_c = \frac{\sum m_i \dot{r}_i}{M}$$

$$T_i = \frac{1}{2} m_i \dot{r}_i \dot{r}_c = \frac{1}{2} m_i (\dot{r}_c + \dot{r}_i')(\dot{r}_c + r_i')$$

$$r_i = r_c + r_i'$$